Supplementary Material: What Do Fairness Mitigations Do to Intersectional Gaps in Cardiovascular Risk Prediction?

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Table S1: Group-wise AUROC and calibration-in-the-large (mean predicted minus observed prevalence); groups: younger F, younger M, older F, older M. Mean over 20 seeds per scenario.

	AUROC				CITL			
Model	yF	yM	oF	oM	yF	yM	oF	oM
Natural								
FedAvg	0.80	0.79	0.72	0.70	−0.003	+0.009	+0.005	−0.004
Naive penalty λ=8	0.78	0.76	0.67	0.65	+0.065	+0.061	−0.004	−0.084
Oversample λ=8	0.78	0.77	0.68	0.65	+0.381	+0.386	+0.363	+0.293
Capped								
FedAvg	0.79	0.80	0.72	0.70	−0.001	+0.011	+0.007	−0.005
Naive penalty λ=8	0.77	0.76	0.67	0.65	+0.067	+0.062	−0.004	−0.083
Oversample λ=8	0.77	0.76	0.67	0.65	+0.363	+0.364	+0.312	+0.237

This supplement contains the additional tables, robustness checks, MLP results, convergence checks, the exponentiated-gradient reference and the UCI replication referred to in the main paper. Section letters and table numbers (S1, S2, ...) match the references in the main text.

A Additional tables

Exploratory and per-λ tables referred to in the main text.

B Additional robustness checks

Client size and age cut-off. The oversampling EO effect appears at every client size (EO $+0.029$ $[+0.004, +0.052]$ at n=75, $+0.032$ $[0.000, +0.062]$ at 150, $+0.040$ $[+0.014, +0.063]$ at 300, $+0.071$ natural), so it is not an artifact of tiny cells. It is absent for an age cut-off of 45 (EO $+0.003$ $[-0.005, +0.011]$ natural, $+0.000$ $[-0.007, +0.007]$ capped) and present for 65 ($+0.096$, $+0.056$) and for three age bands ($+0.090$; capped $+0.027$ $[-0.001, +0.055]$) (Table S5). The offset rises with the cut-off in both settings, which fits the larger prevalence spread at higher cut-offs, but this is three points and we claim no mechanism.

Shrinkage gives no benefit. Offsets are 0.000 in natural data (every client–group cell is large, so shrinkage is numerically the naive penalty) and, in capped data, DP -0.002 $[-0.003, -0.001]$ with EO $+0.005$ $[+0.003, +0.007]$ (leave-states-out: $+0.006$ $[+0.003, +0.008]$): a statistically detectable but practically negligible *worsening* of EO. We report it as no benefit, not as a null.

C A trained MLP (exploratory)

The MLP (16 tanh units) uses T=400 rounds at rate 0.1 and reaches the same unpenalized AUROC as logistic regression (0.797 natural, 0.795 capped). Penalized MLPs are the least reliable models in this study. In a single-seed check (capped, seed 42, λ=8) the intersectional DP gap was 0.091, 0.125 and 0.135 after 50, 150 and 400 rounds at rate 0.1–0.25, and EO_{int} 0.074, 0.116 and 0.133, so the earlier T=50 MLP overstated what the penalty achieves; at T=400 a rate of 0.25 instead of 0.1 still gave EO_{int} 0.208 against 0.133 at nearly identical AUROC (0.686 vs. 0.687). We therefore treat all MLP results as exploratory and do not draw conclusions from small differences.

With that caveat (Tables S6, S7): the sex-only MLP shows the same masking pattern as logistic regression (natural: sex DP $0.125 \to 0.045$, -64%, but DP_{int} $0.266 \to 0.196$, -26%; capped DP_{int} $0.260 \to 0.190$, -27%). The intersectional penalty is *less* effective on DP for the MLP than for logistic regression (DP_{int} 0.087 natural, 0.096 capped at λ=8, against 0.033 and 0.048), but its EO gap is lower (0.164 and 0.157). Oversampling lowers DP_{int} more than for logistic regression (-0.078 $[-0.102, -0.045]$ natural, 5 seeds; -0.039 $[-0.061, -0.018]$ capped, 10 seeds). Its EO effect is positive in natural data ($+0.048$ $[-0.005, +0.096]$; label balancing $+0.056$ $[+0.021, +0.098]$) and null in capped data ($+0.003$ $[-0.054, +0.047]$). None of the 12 MLP primary tests is significant after Holm correction: natural has 5 seeds, for which the smallest attainable p is 0.0625, and the capped adjusted values are at least 0.094. Group re-weighting lowers EO in natural data (-0.013 $[-0.020, -0.008]$) and not detectably in capped data. The MLP therefore neither confirms nor contradicts the logistic-regression EO penalty of oversampling; it suggests that the DP gain from oversampling is not specific to logistic regression.

D Convergence of penalized models

For each penalty type and $\lambda \in \{1, 8, 16\}$ we compared the production schedule of Sec. III-A with half the learning rate and twice the rounds, without the divergence guard (Table S10; absolute differences, mean over settings; capped seeds 42–46 for all arms; natural seeds 42–46 for intersectional, oversampling,

Table S2: Offsets at each λ separately (rule (q)): naive frontier AUROC and EO at that λ, then offsets at the method's own AUROC ('–' if fewer than half of the seeds fall inside the naive AUROC range: capped λ=0.5, 3 of 20 seeds). Natural: naive frontier from the dense grid $\lambda \in \{0.5, 1, 2, 4, 8, 16\}$, 20 seeds; capped: core grid. Mean [95% CI].

	Naive	Naive	Oversample offset		Group re-weight offset	
Setting, λ	AUROC	EO_{int}	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Nat. 0.5	0.751	0.100	−0.011 [−0.013, −0.008]	+0.060 [+0.051, +0.069]	–	–
Nat. 1	0.717	0.097	−0.018 [−0.021, −0.016]	+0.085 [+0.076, +0.095]	+0.003 [+0.002, +0.003]	−0.023 [−0.030, −0.017]
Nat. 2	0.702	0.193	−0.015 [−0.018, −0.013]	+0.071 [+0.065, +0.077]	+0.002 [+0.002, +0.003]	−0.008 [−0.012, −0.005]
Nat. 4	0.688	0.212	−0.012 [−0.014, −0.009]	+0.057 [+0.052, +0.062]	+0.001 [+0.001, +0.002]	−0.006 [−0.008, −0.004]
Nat. 8	0.680	0.222	−0.009 [−0.011, −0.007]	+0.046 [+0.040, +0.053]	+0.001 [+0.000, +0.001]	−0.007 [−0.009, −0.005]
Nat. 16	0.676	0.229	−0.007 [−0.008, −0.005]	+0.038 [+0.031, +0.046]	–	–
Cap. 0.5	0.751	0.129	–	–	–	–
Cap. 1	0.719	0.126	−0.002 [−0.010, +0.006]	+0.001 [−0.019, +0.021]	−0.000 [−0.002, +0.002]	−0.007 [−0.014, +0.000]
Cap. 2	0.704	0.148	−0.007 [−0.015, +0.001]	−0.002 [−0.024, +0.019]	+0.002 [+0.001, +0.004]	−0.008 [−0.015, −0.001]
Cap. 4	0.691	0.165	−0.010 [−0.019, −0.001]	+0.020 [−0.007, +0.047]	+0.002 [+0.001, +0.004]	−0.011 [−0.018, −0.005]
Cap. 8	0.681	0.171	−0.007 [−0.016, +0.003]	+0.037 [+0.003, +0.070]	+0.002 [+0.001, +0.004]	−0.010 [−0.016, −0.005]
Cap. 16	0.676	0.160	−0.008 [−0.018, +0.003]	+0.037 [−0.001, +0.073]	+0.001 [−0.000, +0.003]	−0.007 [−0.014, +0.001]

Table S3: Control experiments for the oversampling EO effect (exploratory): offsets vs. the dense naive frontier at equal AUROC, mean [95% CI over 20 seeds per scenario]. Upper rows use the core λ points for every arm; the last three rows use the dense natural grid ($\lambda \in \{0.5, \ldots, 16\}$).

	Natural (20 seeds)		Capped (20 seeds)	
Variant	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Oversample (cells, noise)	−0.011 [−0.013, −0.009]	+0.056 [+0.051, +0.061]	−0.007 [−0.015, +0.001]	+0.024 [−0.000, +0.047]
orig. n_k in FedAvg	−0.011 [−0.013, −0.009]	+0.056 [+0.051, +0.061]	−0.007 [−0.015, +0.001]	+0.025 [+0.000, +0.048]
no noise (pure duplication)	−0.011 [−0.013, −0.009]	+0.056 [+0.052, +0.061]	−0.008 [−0.015, +0.000]	+0.027 [+0.005, +0.047]
Label-balance by duplication only	−0.004 [−0.005, −0.002]	+0.042 [+0.038, +0.046]	−0.008 [−0.015, −0.002]	+0.020 [−0.008, +0.045]
Loss reweight: labels only	−0.006 [−0.007, −0.006]	+0.030 [+0.026, +0.034]	−0.007 [−0.009, −0.004]	+0.012 [+0.005, +0.021]
Loss reweight: (group,label) cells	−0.003 [−0.003, −0.002]	+0.027 [+0.024, +0.031]	−0.006 [−0.012, −0.000]	+0.020 [+0.004, +0.034]
Loss reweight: groups only	+0.001 [+0.001, +0.002]	−0.012 [−0.014, −0.009]	+0.002 [+0.000, +0.003]	−0.010 [−0.016, −0.005]
Oversample, dense natural grid	−0.012 [−0.014, −0.010]	+0.059 [+0.054, +0.064]	–	–
Label-balance, dense natural grid	−0.002 [−0.003, −0.001]	+0.029 [+0.026, +0.032]	–	–
Loss reweight: groups only, dense natural grid	+0.002 [+0.001, +0.002]	−0.011 [−0.013, −0.009]	–	–

label-balance and group re-weighting at $\lambda \in \{8, 16\}$ and for the EO penalty at $\lambda \in \{1, 8\}$). In natural data, the setting that carries the headline oversampling result, the schedule does not matter: the largest EO difference over 40 DP-type and 10 EO-penalty comparisons is 0.005 and 0.007, an order of magnitude below the effect sizes (0.03–0.07). The sex-only, age-only and marginal-both arms were checked in capped data only. For the DP-type penalties the mean differences are small (capped: AUROC 0.002, DP 0.003, EO 0.009; oversampling alone 0.006, 0.006 and 0.020, the other DP-type arms 0.001, 0.002 and 0.007) but not negligible per seed: the largest EO difference is 0.111, for oversampling at $\lambda \geq 8$, where the production schedule gave the higher EO. For the EO penalty the schedule matters: unguarded, two fits (seed 45, λ=8 and 16) collapsed to AUROC 0.50, which alone puts the mean differences at 0.03 AUROC and 0.13 in DP and EO (maximum 0.74); without them the means are 0.004 AUROC, 0.019 DP and 0.031 EO, so the EO penalty is more schedule-sensitive than the DP-type penalties (0.003 DP, 0.009 EO). Converged-schedule results for the EO penalty at $\lambda \geq 8$ are therefore indicative only.

E Exponentiated-gradient reference

We ran an exponentiated-gradient reduction [1] with class-balanced error (plain 0–1 error collapses to an all-negative classifier at 9.3% prevalence), DP or EO constraints on the four groups and $\epsilon \in \{0.05, 0.02, 0.01\}$, centralized on pooled training data (30K-row subsample in the natural setting; class balancing by resampling for Fairlearn, by weights for our own implementation). We report two implementations, our own and Fairlearn 0.15 (run by the authors externally). Both output a randomized classifier, and we report its expected rates. The per-ϵ results (24 rows; released with the code) are illustrative only: the objective is class-balanced, so the classifier selects about 38–44% of respondents (prevalence 9.3%) and operates far from the 9.3% rule used elsewhere, and its AUROC is that of the mixture score. The two implementations disagree (natural, DP constraint at ϵ=0.01: EO_{int} 0.118 versus 0.210). We did not investigate the cause; the oracle, the number of iterations, the Lagrange-multiplier bound and class balancing by weights versus resampling all differ, and a randomized mixture is sensitive to them. EG numbers should therefore not be used as a benchmark here. We draw no

Table S4: Leave-states-out, deployable threshold: offsets vs. the naive penalty on the core-grid chord (logistic regression, 10 natural / 20 capped partitions), mean [95% CI]. Exploratory.

	Natural		Capped	
Variant	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Oversample	−0.025 [−0.027, −0.021]	+0.072 [+0.068, +0.077]	−0.002 [−0.008, +0.004]	+0.010 [−0.015, +0.036]
Label-balance (dup.)	−0.012 [−0.015, −0.009]	+0.050 [+0.046, +0.054]	−0.006 [−0.011, −0.001]	+0.017 [−0.007, +0.042]
Loss re-weight: labels	−0.012 [−0.013, −0.011]	+0.038 [+0.033, +0.044]	−0.008 [−0.010, −0.005]	+0.021 [+0.010, +0.034]
Loss re-weight: cells	−0.010 [−0.012, −0.009]	+0.037 [+0.030, +0.044]	−0.009 [−0.013, −0.006]	+0.013 [−0.004, +0.031]
Loss re-weight: groups	+0.002 [+0.001, +0.002]	−0.006 [−0.008, −0.003]	+0.002 [−0.000, +0.004]	−0.005 [−0.013, +0.004]
Shrinkage	−0.000 [−0.000, −0.000]	+0.000 [−0.000, +0.000]	−0.002 [−0.002, −0.001]	+0.006 [+0.003, +0.008]

Table S5: Age cut-offs (exploratory). Offsets vs. the naive penalty of the same configuration at equal AUROC, mean [95% CI].

	Natural (10 seeds)		Capped (20 seeds)	
Variant	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Age cut 45: oversample	−0.005 [−0.007, −0.003]	+0.003 [−0.005, +0.011]	−0.002 [−0.006, +0.001]	+0.000 [−0.007, +0.007]
Age cut 45: shrinkage	−0.000 [−0.000, +0.000]	−0.000 [−0.000, −0.000]	−0.001 [−0.002, −0.001]	−0.001 [−0.004, +0.001]
Age cut 65: oversample	−0.037 [−0.041, −0.034]	+0.096 [+0.090, +0.103]	−0.014 [−0.024, −0.002]	+0.056 [+0.026, +0.086]
Age cut 65: shrinkage	−0.000 [−0.000, +0.000]	+0.000 [−0.000, +0.000]	−0.001 [−0.002, +0.001]	+0.005 [+0.002, +0.008]
3 age bands × sex: oversample	+0.012 [+0.009, +0.014]	+0.090 [+0.074, +0.107]	−0.012 [−0.018, −0.005]	+0.027 [−0.001, +0.055]

Table S6: MLP (16 tanh units, T=400, rate 0.1) at λ=8; mean over seeds (5 natural, 10 capped). Exploratory.

Method	AUROC	DP_{sex}	DP_{int}	EO_{sex}	EO_{int}
Natural (5 seeds)					
FedAvg	0.797	0.125	0.266	0.312	0.477
Sex-only	0.791	0.045	0.196	0.096	0.365
Naive inter.	0.693	0.020	0.087	0.036	0.164
Oversample	0.712	0.012	0.042	0.029	0.228
Group re-weight	0.691	0.022	0.083	0.043	0.141
Capped (10 seeds)					
FedAvg	0.795	0.117	0.260	0.285	0.463
Sex-only	0.787	0.039	0.190	0.083	0.354
Naive inter.	0.688	0.027	0.096	0.057	0.157
Oversample	0.703	0.023	0.079	0.063	0.161
Group re-weight	0.687	0.027	0.093	0.056	0.156

comparison with the other methods.

F UCI Heart Disease (secondary, 851 patients)

On UCI (four hospitals; stratified 5-fold CV repeated 20 times; rule (q); bootstrap CIs over patients; the same stiffness-aware training schedule and guard as for BRFSS) the sex-only model at λ=8 reduces sex DP and EO by 63% and 75% but age×sex gaps by 32% and 33% (FedAvg AUROC 0.878, sex-only 0.852). Oversampling offsets are not reported: its AUROC at λ=8 (0.736) is far below the naive penalty's (0.829), so offsets would rest on two frontier points. With subgroup cells of 2–6 patients per client, UCI is too small to discriminate mechanisms; its intervals omit training variability.

References

[1] A. Agarwal, A. Beygelzimer, M. Dudík, J. Langford, and H. Wallach, "A reductions approach to fair classification," in *Proc. ICML*, PMLR vol. 80, 2018, pp. 60–69.

Table S7: MLP frontier offsets vs. naive penalty (rule (q)), mean [95% CI over seeds]; 5 natural, 10 capped seeds. Exploratory.

Method	Natural DP_{int}	Natural EO_{int}	Capped DP_{int}	Capped EO_{int}
Oversample	−0.078 [−0.102, −0.045]	+0.048 [−0.005, +0.096]	−0.039 [−0.061, −0.018]	+0.003 [−0.054, +0.047]
Label-balance (dup.)	−0.049 [−0.077, −0.019]	+0.056 [+0.021, +0.098]	−0.042 [−0.062, −0.022]	+0.019 [−0.034, +0.066]
Group re-weight	−0.002 [−0.004, +0.001]	−0.013 [−0.020, −0.008]	−0.003 [−0.006, +0.000]	−0.005 [−0.013, +0.002]

Table S8: Fine penalty grid, $\lambda \in \{0.1, \ldots, 16\}$ (logistic regression, rule (q); mean over 20 capped / 5 natural seeds). EO of the intersectional and marginal-both penalties is U-shaped; sex-only stays high for all λ.

	Intersectional			Marginal-both			Sex-only		
λ	AUROC	DP_{int}	EO_{int}	AUROC	DP_{int}	EO_{int}	AUROC	DP_{int}	EO_{int}
Capped (20 seeds)									
0.1	0.789	0.222	0.352	0.778	0.202	0.273	0.794	0.201	0.378
0.25	0.774	0.191	0.249	0.751	0.139	0.131	0.792	0.186	0.354
0.5	0.751	0.138	0.129	0.719	0.083	0.127	0.790	0.178	0.341
1	0.719	0.084	0.126	0.703	0.059	0.154	0.789	0.174	0.333
2	0.704	0.062	0.148	0.689	0.048	0.177	0.788	0.171	0.329
4	0.691	0.053	0.165	0.680	0.042	0.194	0.788	0.169	0.329
8	0.681	0.048	0.171	0.676	0.043	0.182	0.787	0.167	0.325
16	0.676	0.049	0.160	0.673	0.044	0.172	0.786	0.164	0.312
Natural (5 seeds)									
0.1	0.791	0.228	0.371	0.781	0.210	0.294	0.795	0.200	0.375
0.25	0.777	0.199	0.272	0.753	0.122	0.100	0.793	0.186	0.354
0.5	0.752	0.122	0.100	0.719	0.079	0.094	0.791	0.179	0.344
1	0.719	0.076	0.105	0.703	0.046	0.193	0.790	0.175	0.337
2	0.703	0.046	0.195	0.689	0.038	0.217	0.790	0.172	0.335
4	0.689	0.037	0.218	0.681	0.033	0.227	0.789	0.171	0.333
8	0.681	0.032	0.228	0.677	0.031	0.232	0.789	0.171	0.333
16	0.677	0.030	0.235	0.674	0.030	0.234	0.789	0.171	0.333

Table S9: Group re-weighting with local vs. global group counts under Dirichlet partitions (federated, 10 seeds): offsets vs. naive penalty, mean [95% CI]. Exploratory.

Partition	Weights	DP_{int}	EO_{int}
Dirichlet α=1	local group counts	−0.005 [−0.007, −0.003]	+0.016 [+0.008, +0.026]
Dirichlet α=1	global group counts	+0.001 [+0.001, +0.002]	−0.003 [−0.005, −0.001]
Dirichlet α=0.3	local group counts	−0.007 [−0.014, −0.000]	+0.027 [+0.009, +0.047]
Dirichlet α=0.3	global group counts	+0.002 [+0.001, +0.002]	−0.008 [−0.012, −0.005]

Table S10: Production schedule vs. half learning rate / double rounds (unguarded): mean absolute difference in AUROC, DP_{int}, EO_{int}; n = settings; last column: largest EO difference.

Scenario	Penalty	n	$\overline{\lvert\Delta\rvert}$ AUROC	$\overline{\lvert\Delta\rvert}$ DP_{int}	$\overline{\lvert\Delta\rvert}$ EO_{int}	max $\lvert\Delta\rvert$ EO_{int}
capped	DP	90	0.0016	0.0028	0.0089	0.111
capped	EO (excl. collapsed fit)	13	0.0035	0.0188	0.0308	0.070
capped	EO	15	0.0310	0.1284	0.1256	0.743
natural	DP	40	0.0002	0.0004	0.0009	0.005
natural	EO	10	0.0001	0.0006	0.0019	0.007
capped	DP: oversample	15	0.0061	0.0056	0.0196	0.111
capped	DP: other DP arms	75	0.0007	0.0022	0.0067	0.039
natural	DP: oversample	10	0.0004	0.0003	0.0011	0.005
natural	DP: other DP arms	30	0.0001	0.0004	0.0009	0.005